

**Jomo Kenyatta University of Agriculture and
Technology**

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**A Causal Evaluation of Culturally
Adaptive AI
as Relational Scaffolding for Loneliness
Intervention**

A Research Proposal Submitted in Partial Fulfilment
of the Requirements for the Degree of
Master of Science in Artificial Intelligence

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Abstract

Digital mental health interventions show considerable promise for tackling the global loneliness epidemic, yet conversational AI systems consistently underserve non-Western populations. Built on Western individualistic psychiatric frameworks, these models carry severe epistemic bias: they routinely misread communal expressions of psychological distress. For instance, when a user in Nairobi describes *kufikiria sana* (thinking too much), standard models tend to pathologise this as clinical anxiety rather than recognising the communal disruption it signifies. Furthermore, many commercial platforms foster what may be termed the “Companionship Paradox,” functioning as synthetic replacements for human connection instead of temporary therapeutic scaffolding aimed at restoring real-world relationships.

Network analysis of adolescent psychosocial functioning in Kenya identifies loneliness as the most central psychological variable, bridging depressive symptoms and diminished well-being through gender-contingent pathways: family support proves more central for girls, whereas peer support plays a larger role for boys (Ndetei et al., 2025). This research proposes a culturally adaptive conversational AI built on a dual-track Retrieval-Augmented Generation (RAG) pipeline that grounds the intervention in both East African cultural knowledge and evidence-based Cognitive Behavioural Therapy (CBT). The system translates standard behavioural activation protocols into culturally resonant metaphors while prioritising relational well-being and community reintegration over isolated symptom management.

To establish clinical efficacy, this study employs a rigorous causal inference framework. Current human-computer interaction (HCI) research suffers from over-reliance on correlational engagement metrics. By deploying structural causal models alongside deep graph learning techniques, this work systematically compares culturally adapted outputs against generic baseline responses, isolating the therapeutic “outcome delta.” This methodological advance addresses a critical question: does algorithmic cultural adaptation directly and causally drive longitudinal loneliness reduction? The resulting open-source framework will offer a scalable blueprint for ethical digital health infrastructure serving diverse populations globally.

1. Problem Formulation and Research Objectives

1.1 Statement of the Problem and Research Gaps

The rapid integration of generative AI into digital mental health has created new possibilities for scalable loneliness intervention. However, deployment in resource-constrained, non-Western settings reveals critical structural and methodological shortcomings. Three primary research gaps constrain current paradigms:

Cultural Inflexibility and Epistemic Bias.

Existing AI companions fundamentally lack adaptation for non-Western linguistic and social contexts. Foundation models, trained predominantly on Western corpora, misinterpret communal idioms of distress with alarming regularity. Systematic reviews of conversational agents for mental health have found widespread reliance on Western-developed scales, minimal cultural adaptation, a predominance of small and short-duration studies, and weak links between automated performance metrics and actual user well-being ([Evaluating Text-based Conversational Agents, 2026](#)). Ethnographic research in urban Kenya has documented *kufikiria sana* as a powerful idiom that reflects both individual suffering and social anxieties, embedding psychological distress firmly within relational contexts ([Mendenhall et al., 2019](#)). These findings demand intervention designs that validate, rather than pathologise, culturally specific expressions.

The Absence of Relational Scaffolding.

Most commercial conversational agents optimise for prolonged user engagement rather than clinical recovery. Recent longitudinal research reveals a troubling paradox: AI companions provide unconditional, unflagging support that attracts socially struggling individuals, yet extended use may erode well-being and impair the capacity to navigate real-world relationships ([Jovanovic, 2026](#)). Investigations into the relationship between depression and conversational AI use among college students have found that depression positively associates with AI companion use, with loneliness mediating this link ([Depression and Conversational AI, 2025](#)). A critical gap remains in designing systems that deliver evidence-based therapeutic responses while actively guiding users toward renewed human community engagement.

The HCI Evaluation Deficit.

Evaluations of mental health chatbots rely heavily on weak, correlational data. Researchers frequently fail to isolate precise mechanisms of clinical change. Systematic reviews and meta-analyses of generative AI chatbots for mental health have underscored the urgent need for large-scale, multicultural trials with culturally adapted AI to optimise safety, efficacy, and equity ([Viduani et al., 2025](#)). A rigorous causal inference framework is needed to quantify the true impact of localised conversational interventions.

1.2 Aim of the Study

The overarching aim is to design, implement, and evaluate personalised AI conversations that measurably reduce loneliness. This study will engineer a culturally adaptive generative pipeline using Microsoft Azure OpenAI and a RAG architecture. By grounding the model in both evidence-based CBT and localised cultural knowledge, the system will translate standard therapeutic strategies into culturally resonant dialogue. Through structural comparison of localised interventions against generic baseline re-

sponses using causal models, this research will isolate the therapeutic “outcome delta,” demonstrating the clinical necessity of deep algorithmic cultural adaptation.

1.3 Research Questions

Primary Research Question:

How can a personalised, culturally adaptive conversational AI companion be designed, fine-tuned, and rigorously evaluated to deliver evidence-based interventions that measurably reduce loneliness in target populations?

Secondary Research Questions:

RQ2a (Architectural): How does anchoring generative models within a localised RAG pipeline mitigate the epistemic bias and clinical hallucination present in standard Western-trained models?

RQ2b (Evaluative): What is the quantifiable, causal impact of algorithmic cultural adaptation on longitudinal loneliness reduction when structural causal models compare culturally tailored responses against generic therapeutic baselines?

2. Literature Review

Integrating large language models (LLMs) into digital mental health creates opportunities for scalable loneliness support. However, deploying these systems in non-Western environments exposes significant limitations. This review synthesises recent work on conversational agents, epistemic bias, cultural adaptation, and causal evaluation.

2.1 Efficacy and Limitations of Generic Conversational Agents

Rule-based and generative AI chatbots have demonstrated capacity for delivering structured psycho-education and CBT prompting. Meta-analytic evidence suggests that generative AI mental health chatbots outperform rule-based systems in reducing mental health symptoms (Viduanı et al., 2025). Evaluations of LLMs delivering CBT indicate that, while these models generate plausible CBT-like dialogues, they remain limited in conveying empathy and maintaining therapeutic consistency (Bedi et al., 2026). Comparative studies of generalist versus specialised models in RAG-based mental health dialogue have found that generalist reasoners surpass domain-specific fine-tuned models in empathy, suggesting that reasoning capability matters more than domain-specific vocabulary (Kafi et al., 2026).

Despite these technical capabilities, the clinical value of purely generative models remains contested. A cross-cultural study spanning 4,641 participants across seven countries revealed mixed outcomes: greater short-term comfort co-occurred with increases in language about loneliness, depression, and suicidal ideation over time (Amat-Lefort et al., 2026). This “Companionship Paradox” suggests that highly responsive LLMs

risk functioning as “relational mirages,” where sustained reliance on AI for emotional regulation can paradoxically deepen offline isolation.

2.2 Cultural Adaptation and Epistemic Constraints

Deploying conversational interventions in Sub-Saharan Africa exposes severe epistemic biases embedded in foundation models. Research on multilingual depression classification has shown that model accuracy varies markedly across languages, with demographic biases affecting performance in non-English settings. [Xie \(2025\)](#) developed a novel approach for dynamic cultural adaptation of LLM-based mental health agents through context engineering for Chinese American caregivers. The culturally adapted agent produced significant improvements in near-term emotional well-being and received higher ratings for cultural competence and therapeutic alliance relative to a non-adapted baseline.

This line of inquiry has extended to additional populations. Researchers developed Molhim, a culturally adapted multimodal conversational AI platform for PTSD screening, demonstrating the feasibility of structured clinical dialogue across cultural boundaries ([Ozel et al., 2026](#)). An Ubuntu-guided LLM framework integrating CBT with the African philosophy of Ubuntu has been proposed as a proof of concept for culturally sensitive AI mental health dialogue ([An Ubuntu-Guided LLM Framework, 2026](#)). Particularly relevant to East Africa, the Shamiri Institute has launched shamiriAI, an AI-powered mental health platform for youth across Africa, building on a community-based model that has reached over 185,000 young people in Kenya through school-based interventions led by trained peer providers ([Shamiri Institute, 2025](#)).

2.3 Idioms of Distress in East African Contexts

Psychological distress in East African populations often manifests through culturally specific idioms that do not align with Western diagnostic categories. Analysis of 100 life-history narrative interviews among patients at a public hospital in Nairobi revealed that *kufikiria sana* emerged organically as a salient cultural idiom that functions both as a standalone expression and as a marker of other psychological distress ([Mendenhall et al., 2019](#)). This ethnopsychology integrates local (traditional) and biomedical frameworks that are both distinct and overlapping in their symptomatology and experiential dimensions.

Network-based analysis of adolescent psychosocial functioning in Kenya has identified loneliness as the most central psychological variable, directly bridging depressive symptoms and diminished well-being ([Ndeti et al., 2025](#)). Family support proves more central for girls, whereas peer support plays a greater role for boys, underscoring the need for contextually and gender-sensitive interventions. The Swahili expression *ipo siku* (there will come a day) and the colloquial term “Njaanuary” reflect local coping mechanisms and cultural understandings of hardship that must inform any culturally competent intervention.

2.4 Causal Inference in Mental Health AI

Current evaluation of HCI in mental health relies largely on correlational longitudinal studies. Recent methodological advances offer pathways to address this limitation. Causal graph neural networks have been proposed as a framework for integrating causal inference with graph-based deep learning, enabling a shift from pattern recognition to causal reasoning in healthcare AI (Mesinovic et al., 2026). InsightFlow demonstrated that LLMs can produce clinically meaningful causal graphs from patient-therapist dialogues, achieving structural similarity comparable to inter-annotator agreement (Gupta et al., 2026).

Wang et al. (2025) identified daily-living features related to loneliness using causal machine learning, establishing that social interactions, physical activity, sleep, and cardiac features have a causal influence on loneliness. A Bayesian decision-theoretic framework for ranking digital mental health interventions based on individual presentation has also been proposed (Causal AI Recommendation System, 2025). Nevertheless, no existing study has extended causal inference approaches to evaluate cultural adaptation in conversational AI.

2.5 Research Gap Synthesis

The reviewed literature reveals a clear gap: while RAG-based cultural adaptation has shown promise in Chinese American and other non-Western contexts, and while East African idioms of distress have been extensively documented, no study has combined these insights to develop a culturally adapted conversational AI for Kenyan youth. Furthermore, no research has employed causal inference frameworks to isolate the specific effect of algorithmic cultural adaptation on loneliness reduction. This research directly addresses both gaps.

3. Methodology

This research follows a four-phase sequential pipeline designed to capture signals, translate culture, generate relational scaffolding, and causally evaluate outcomes over a six-month period. Phases overlap where feasible to maximise efficiency.

3.1 Phase 1: Signal Ingestion and Clinical Governance (Month 1)

The intervention begins with system setup and safety framework implementation. The pipeline continuously ingests detection confidence scores from the edge deployment layer. When a user’s state crosses a predefined clinical threshold, the active intervention workflow triggers. Safety remains the absolute priority throughout. Every text input passes through strict content safety filters (Azure AI Content Safety) before processing. If the system detects severe distress or self-harm indicators, it immediately bypasses the generative engine and activates a hard-coded clinical escalation protocol.

3.2 Phase 2: Dual-Track Retrieval-Augmented Generation (Months 1–2)

This phase addresses the epistemic bias inherent in global language models through a dual-track RAG pipeline constructed via Azure AI Search:

Track A (Clinical Grounding)

Queries a curated repository of CBT protocols, behavioural activation strategies, and evidence-based psycho-educational materials validated for adolescent populations.

Track B (Cultural Translation)

Simultaneously queries a knowledge base of East African idioms of distress (including *kufikiria sana*, *kufa moyo*, *ipo siku*), culturally resonant metaphors, communal coping strategies, and validated cultural responses. This repository draws from published ethnographic research (Mendenhall et al., 2019), clinical expertise from local mental health practitioners, and community-based participatory design.

The pipeline fuses these knowledge streams into a single contextualised data packet. This architecture mirrors successful implementations in mental health dialogue systems, with empirical evidence showing that RAG significantly reduces hallucinations and improves contextual relevance compared to generation-only approaches (Kafi et al., 2026).

3.3 Phase 3: Generative Formulation for Relational Scaffolding (Months 2–3)

With context fused, the system proceeds to response formulation. The RAG packet feeds into the generative engine via a dynamic system prompt that explicitly constrains the AI persona to function as relational scaffolding rather than a synthetic companion. The prompt includes:

- A directive to prioritise community reintegration over prolonged engagement;
- Instructions to employ locally resonant metaphors and communal language registers;
- Guidelines for action-oriented suggestions promoting real-world social connection;
- Prohibitions against encouraging dependency.

The ultimate goal of every generated response is to guide the user toward re-engaging with their physical community.

3.4 Phase 4: Longitudinal Telemetry and Causal Evaluation (Months 3–6)

This phase constitutes the core scientific contribution. Rather than relying on weak correlational metrics, evaluation is treated as a rigorous causal inference problem.

Study Design.

A two-arm randomised controlled trial (RCT) with $N \geq 200$ Kenyan youth participants (aged 14–25). Participants are randomised into:

- **Intervention arm** ($n \geq 100$): Culturally adaptive RAG pipeline with dual-track retrieval.
- **Control arm** ($n \geq 100$): Generic baseline using the same LLM without cultural RAG.

Instruments.

Loneliness is measured at baseline (T0), week 4 (T1), and week 8 (T2) using the Revised UCLA Loneliness Scale (Version 3), which demonstrates high internal consistency (Cronbach’s α ranging from .89 to .94) and acceptable test–retest reliability over a one-year period ($r = .73$) (Russell, 1996). Secondary measures include therapeutic alliance, engagement duration, and self-reported community reintegration behaviours.

Causal Analysis.

Structural causal models and deep graph learning techniques are deployed to systematically compare the intervention and control arms. Causal graph neural networks enable integration of causal inference with graph-based deep learning to learn invariant mechanisms rather than spurious correlations (Mesinovic et al., 2026). This approach mathematically isolates the cultural adaptation variable, providing evidence for whether cultural adaptation directly causes a reduction in loneliness scores.

3.5 Technical Architecture Summary

Table 1 summarises the key components of the proposed system.

Table 1: Technical architecture components.

Component	Implementation
Foundation Model	Azure OpenAI GPT-4o
Retrieval Engine	Azure AI Search with hybrid retrieval
Safety Filter	Azure AI Content Safety
Clinical Knowledge Base	Curated CBT protocols, behavioural activation strategies
Cultural Knowledge Base	East African idioms, metaphors, communal coping strategies
Validation Instrument	UCLA Loneliness Scale (Version 3)
Causal Framework	Structural causal models, causal graph neural networks

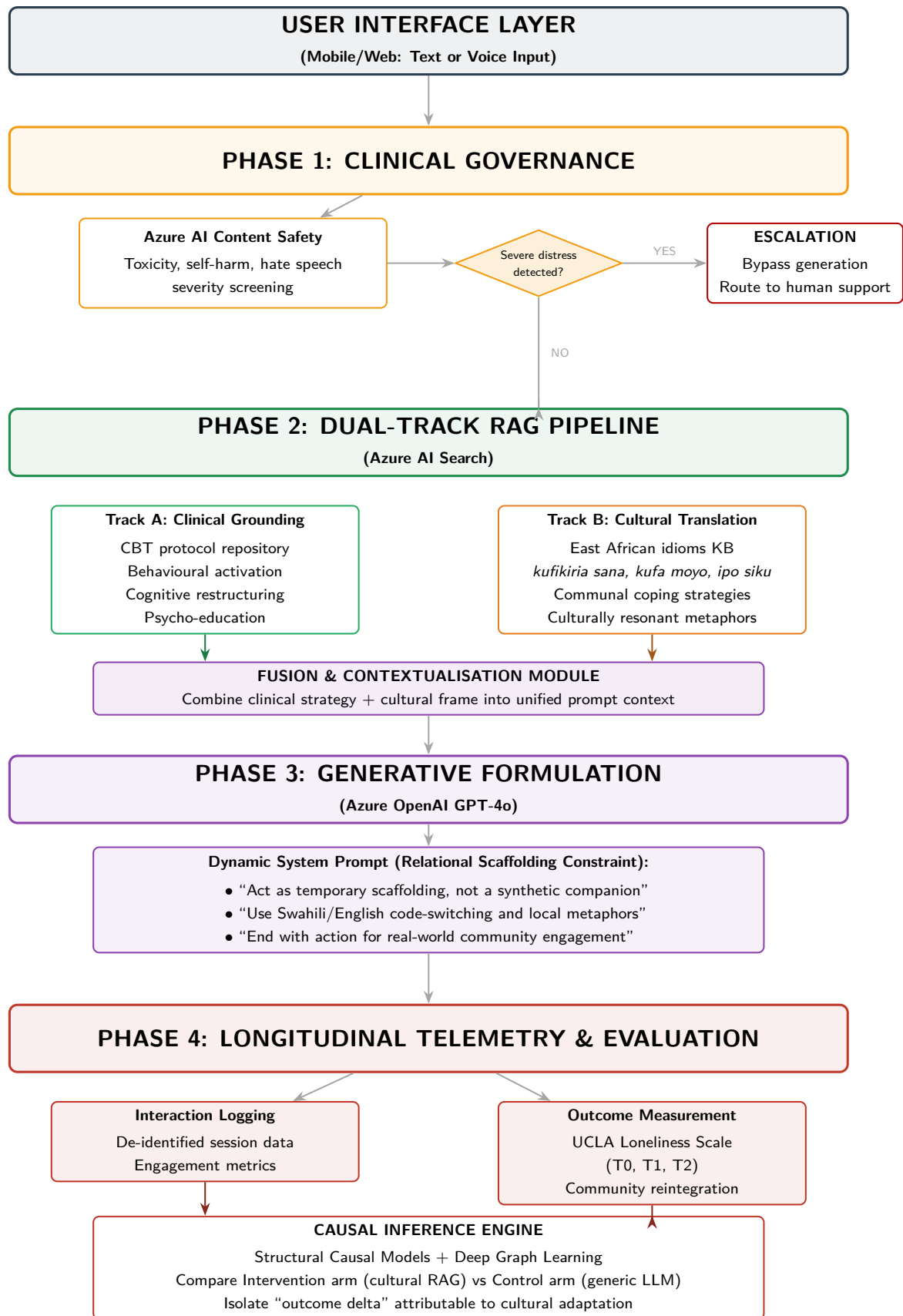


Figure 1: Schematic diagram of the proposed dual-track RAG architecture for culturally adaptive loneliness intervention. The pipeline comprises four sequential phases: (1) Clinical Governance with safety filtering; (2) Dual-Track RAG retrieval from CBT protocols (Track A) and East African cultural knowledge bases (Track B); (3) Generative formulation constrained by relational scaffolding prompts; (4) Longitudinal evaluation using validated psychometric instruments and causal inference models to isolate the effect of cultural adaptation.

4. Timeline (6-Month Project Plan)

Table 2 presents the planned schedule for the research.

Table 2: Project timeline and deliverables.

Phase	Duration	Activities	Deliverables
Phase 1: Preparation	Month 1	Ethical approval (JKUAT + NACOSTI), stakeholder engagement, cultural knowledge base curation, CBT protocol adaptation, system architecture design	Approved ethics, curated KB, community partnerships, architecture specification
Phase 2: Development	Months 1–2	Dual-track RAG pipeline implementation, prompt engineering, safety guardrails, integration testing, pilot testing with $N=10$	Functional prototype, pilot evaluation results
Phase 3: Deployment & RCT	Months 3–5	Full RCT with $N \geq 200$ participants (8-week intervention with T0/T1/T2 UCLA Loneliness Scale administration), interaction logging, longitudinal tracking	Complete dataset, ongoing safety monitoring
Phase 4: Analysis & Write-Up	Months 5–6	Causal inference analysis (structural causal models + deep graph learning), manuscript preparation, open-source release	Thesis, publications, open-source framework

Note: Phases 1 and 2 run concurrently where feasible. Month 1 overlaps preparation and initial development. Months 3–5 include continuous data collection and real-time safety monitoring. Months 5–6 focus on analysis and writing while final data points are collected.

5. Ethical Considerations

This research will comply with JKUAT ethical review guidelines and NACOSTI regulations. Key safeguards include:

Informed Consent

All participants (and guardians for minors) will provide written informed consent following full disclosure of research purpose, procedures, risks, and benefits.

Data Privacy

All interactions will be de-identified, encrypted, and stored on secure servers compliant with Kenyan data protection laws.

Safety Protocol

Detection of severe distress triggers an immediate bypass of generative AI and activates the escalation protocol, providing contact information for human mental health professionals and crisis helplines.

Participant Welfare

No participant will be denied access to alternative mental health services. A licensed clinical psychologist will oversee safety monitoring throughout the study.

Community Engagement

Participatory design workshops with community stakeholders, mental health practitioners, and youth representatives will ensure cultural appropriateness.

Open-Source Commitment

Upon completion, the anonymised dataset, codebase, and knowledge base will be released under an open-source licence.

6. Expected Contributions and Significance

This research makes three primary contributions:

Theoretical Contribution

The first empirical demonstration of a causal link between algorithmic cultural adaptation and loneliness reduction, advancing HCI evaluation methodology beyond correlational approaches.

Technical Contribution

The dual-track RAG architecture, grounded in East African cultural knowledge and CBT, provides an open-source blueprint for culturally adaptive mental health AI applicable to other non-Western contexts. This work builds on established cultural adaptation workflows (Xie, 2025) and extends these methods to communal idioms of distress unique to East Africa.

Practical Contribution

The intervention offers a scalable, low-cost tool to address the documented loneliness crisis among Kenyan youth. By explicitly prioritising community reintegration over prolonged engagement, the system mitigates the Companionship Paradox identified in existing AI companions, aligning with community-based mental health models already implemented by the Shamiri Institute across Kenya (Shamiri Institute, 2025).

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